Roll No. ....

TEC–303

B. TECH. (ECE) (THIRD SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2022
NETWORK ANALYSIS AND SYNTHESIS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) (i) For the circuit shown in Fig. 1, determine the value of equivalent capacitance C_{eq} as seen from terminals *a*-*b* : (CO1)

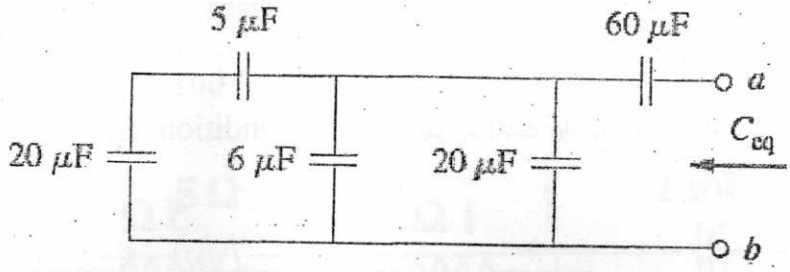

Fig. 1

(ii) For the circuit shown in Fig. 2, determine the value of equivalent capacitance L_{eq}.

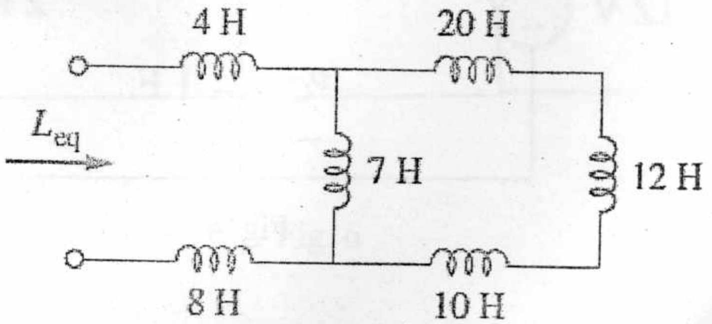

Fig. 2

P. T. O.

OR

(b) Find i_o in the circuit shown in Fig. 3. (CO1)

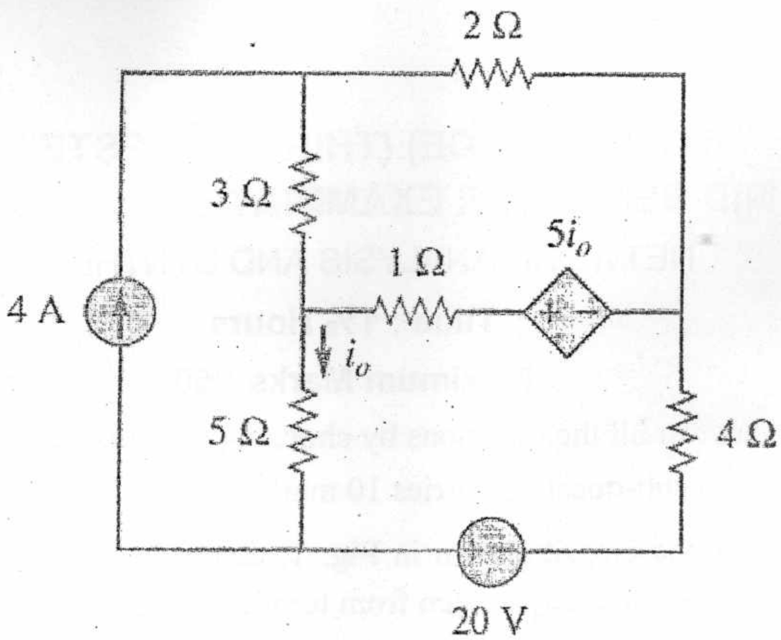

Fig. 3

2. (a) Determine i, v_C and i_L under DC conditions in the circuit of Fig. 4. (CO1)

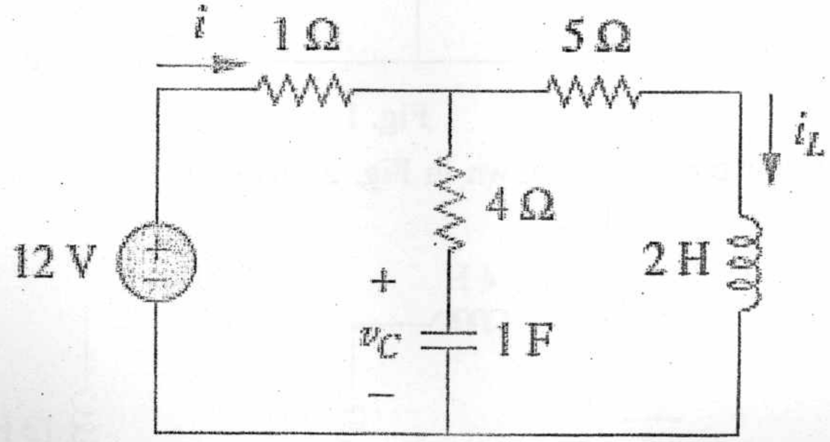

Fig. 4

OR

(b) Under DC conditions, find energy stored in the capacitors as shown in Fig. 5. (CO1)

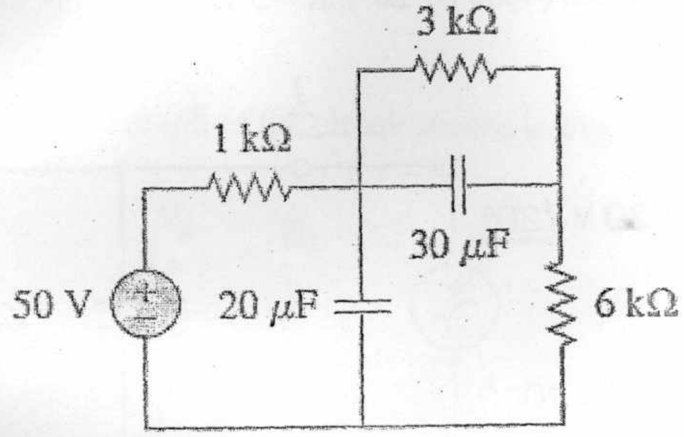

Fig. 5

3. (a) For the circuit shown in Fig. 6, determine Thevenin and Norton equivalent circuit across the load terminals *a-b*.

Given that $v(t) = 100 (\cos (1000\, t)$ volts. (CO1)

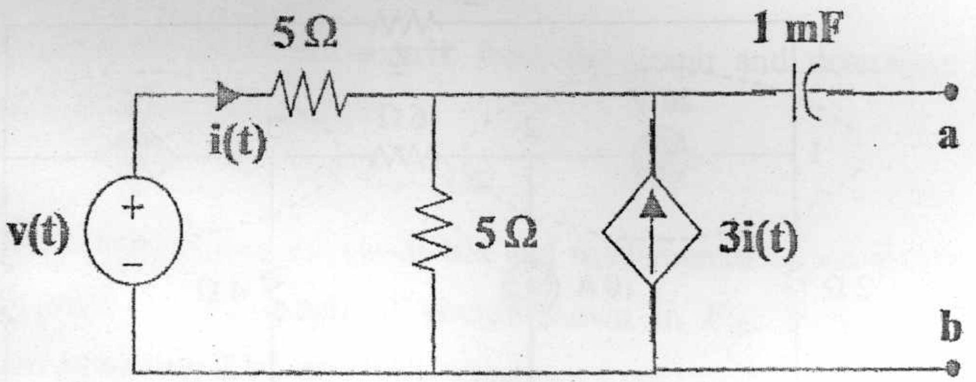

Fig. 6

OR

(b) For the circuit shown in Fig. 7, determine I. Given that $V = 30 \cos(4000\,t + 20°)$, $R = 5\ \Omega$, $C = 25\,\mu F$. (CO1)

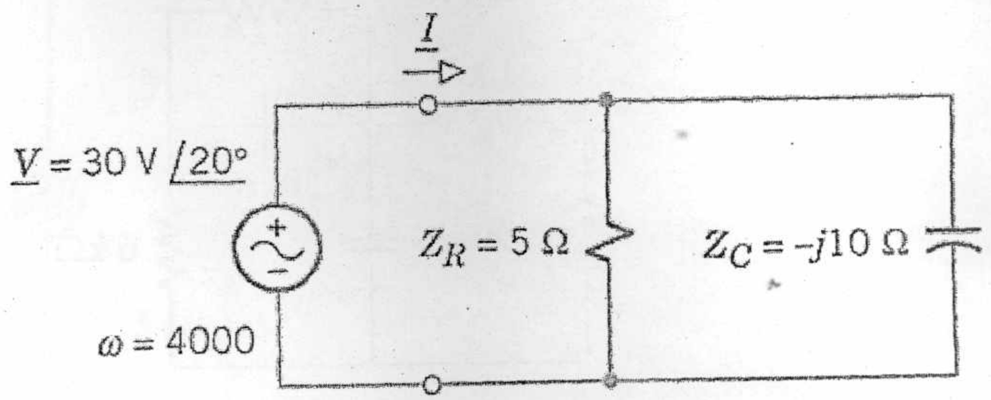

Fig. 7

4. (a) Find the node voltages in the circuit of Fig. 8. (CO1)

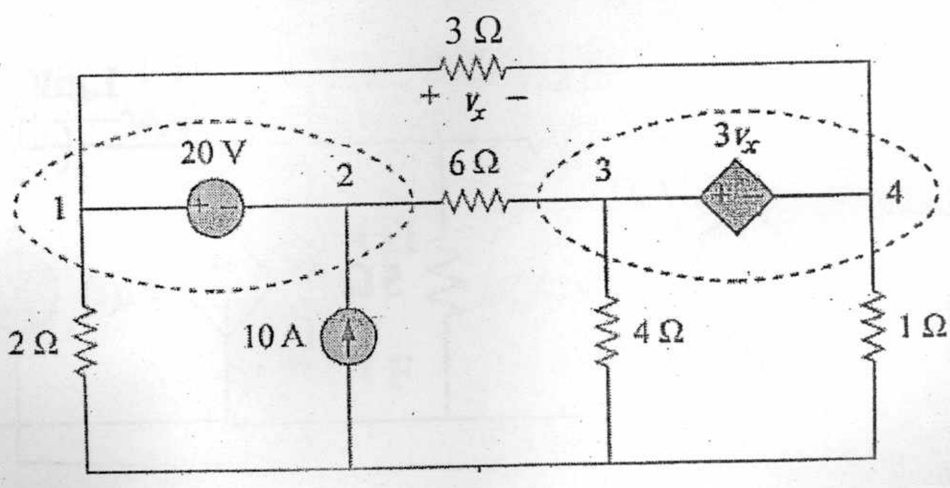

Fig. 8

OR

(b) State and prove maximum power transfer theorem for AC circuits consisting of network sources, resistors, capacitors and inductors. (CO1)

5. (a) Draw the directed graph of the circuit shown in Fig. 9. (CO2)

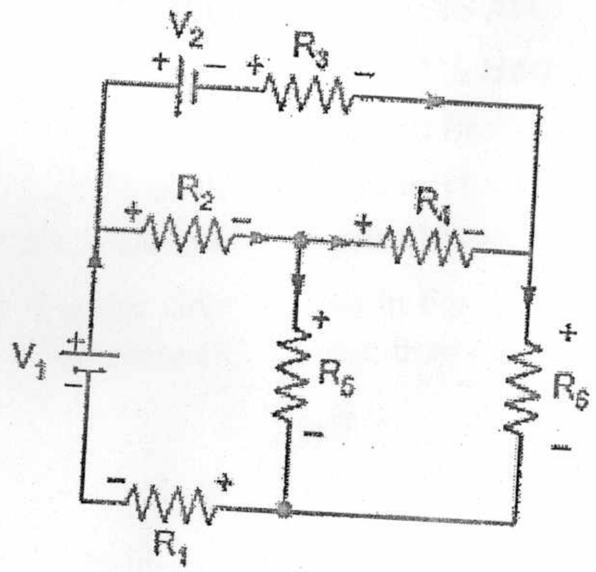

Fig. 9

Deduce the incidence matrix from the graph and determine no. of possible tress of the graph.

OR

(b) Write fundamental cut set matrix and fundamental tie set matrix of any one tree of the graph of circuit shown in Fig. 9 and deduce the corresponding KVL and KCL equations from the matrix. (CO2)